

Multithreaded Chat Application

Project Title

Multithreaded Chat Application

Objective

To build a Java-based multithreaded client-server chat application that allows multiple users to communicate in real-time, using Java Sockets and Multithreading.

Tools & Technologies

- Language: Java
- Concepts Used:
 - Socket Programming
 - Multithreading
 - I/O Streams (BufferedReader, PrintWriter)
 - Collections Framework (HashSet)

Project Structure

ChatApp/

Server.java // Multithreaded server code

Client.java // Chat client code

Server Side: Server.java

- Listens on port 12345
- Accepts multiple client connections
- Uses ClientHandler threads to manage each client
- Broadcasts messages to all connected clients

Client Side: Client.java

- Connects to server via IP and port
- Sends username and messages
- Runs 2 threads:

- One for sending messages
- One for receiving messages

How to Run

1. Compile the files:

```
javac Server.java
```

```
javac Client.java
```

2. Run the Server:

```
java Server
```

3. Run each Client in a new terminal:

```
java Client
```

Output Example

Server: New client connected

Client 1: Rahul joined the chat

Client 2: Aman joined the chat

Client 1: Hello Aman

Client 2: Hi Rahul!

Features

- Real-time messaging
- Handles multiple clients using threads
- Broadcasts messages to all users
- Graceful exit handling
- Easy to run & test

Learning Outcomes

- Understanding Java socket communication

- Implementing multithreaded systems
- Real-world client-server architecture
- Java I/O and networking fundamentals